

# wyvis: A flow-line model of subglacial drainage

## Introduction

**wyvis** is intended as a simple instructive model to help learn about and understand some features of subglacial drainage systems and their effect on ice sliding.<sup>1</sup> It runs in Matlab, and makes use of the in-built solver **ode15s** to integrate the system of stiff ordinary differential equations that arises from discretizing the model along a flow-line.

The model includes a channel and a distributed system, coupled together through an exchange term. If the exchange coefficient is set to zero then the two systems operate independently, so the model can be used to study a single channel, as in the example below. A sliding law relating basal friction to effective pressure can be used to calculate the ice speed, assuming plug flow.

The primary variables are the cross-sectional area  $S$  and hydraulic potential  $\phi_c$  in the channel, and the average sheet depth  $h$  and hydraulic potential  $\phi$  in the distributed system, and the ice speed  $u$ . The equations below should be consulted to understand the physics that is included.

Water inputs from the surface have to be prescribed. These can be fed directly into the channel through a distributed source term  $M_{in}$ , into the sheet through a distributed source term  $m_{in}$ , or into a number of moulins (connected directly to the channel) with source terms  $Q_m$ .

## Equations

Distance along the flow-line is denoted  $x$  and time is denoted  $t$ . There are four main equations expressing the evolution of channel cross-sectional area  $S$  and sheet depth  $h$ , and the conservation of water in channel and sheet,

$$\frac{\partial S}{\partial t} = \frac{\rho_w}{\rho_i} M - \tilde{A} S |N_c|^{n-1} N_c + U_b (h_{rc} - S/\ell_{rc})_+, \quad (1)$$

$$\frac{\partial h}{\partial t} = \frac{\rho_w}{\rho_i} m - \hat{A} S |N|^{n-1} N + U_b (h_r - h)_+ / \ell_r, \quad (2)$$

$$\beta S \frac{\partial \phi_c}{\partial t} + \frac{W \sigma_c}{\rho_w g} \frac{\partial \phi_c}{\partial t} + S_m \frac{\partial \phi_c}{\partial t} \delta(x_m) + \frac{\partial S}{\partial t} + \frac{\partial Q}{\partial x} = M + M_{in} + Q_m \delta(x_m) + \kappa, \quad (3)$$

$$\beta h \frac{\partial \phi}{\partial t} + \frac{W \sigma}{\rho_w g} \frac{\partial \phi}{\partial t} + W \frac{\partial h}{\partial t} + \frac{\partial}{\partial x} (W q) = W m + W m_{in} - \kappa. \quad (4)$$

Hydraulic potential and effective pressure are related by

$$\phi_c = \rho_i g Z_s + (\rho_w - \rho_i) g Z_b - N_c, \quad (5)$$

$$\phi = \rho_i g Z_s + (\rho_w - \rho_i) g Z_b - N. \quad (6)$$

Exchange flow between sheet and channel is

$$\kappa = \lambda(\phi - \phi_c) + \lambda_2 K h^\alpha (\phi - \phi_c). \quad (7)$$

Water flux in channel and sheet are

$$Q = -K_c S^{\alpha_c} \left| \frac{\partial \phi_c}{\partial x} \right|^{-1/2} \frac{\partial \phi_c}{\partial x}, \quad (8)$$

$$q = -K h^\alpha \frac{\partial \phi}{\partial x}. \quad (9)$$

<sup>1</sup>Version2 includes an equation for ice velocity. In the original version ice velocity was prescribed.

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Melting rates are

$$M = \frac{1 - \rho_w c \gamma}{\rho_w L} \left| Q \frac{\partial \phi_c}{\partial x} \right| - \frac{\rho_w g c \gamma}{L} Q \frac{\partial Z_b}{\partial x}, \quad (10)$$

$$m = \frac{G}{\rho_w L} + \frac{\tau_b U_b}{\rho_w L}. \quad (11)$$

Boundary conditions are

$$q = Q = 0 \quad \text{at} \quad x = 0, \quad \phi = \phi_c = \max(0, \rho_w g Z_b) \quad \text{at} \quad x = x_0. \quad (12)$$

The horizontal momentum equation in the ice is

$$RW \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left( 2A^{-1/n} WH \left| \frac{\partial u}{\partial x} \right|^{1/n-1} \frac{\partial u}{\partial x} \right) - \rho_i g WH \frac{\partial Z_s}{\partial x} - W \tau_b(u, N) - HA^{-1/n} \left( \frac{n+2}{2} \right)^{1/n} \left( \frac{u}{W} \right)^{1/n}, \quad (13)$$

where  $H = Z_s - Z_b$  is the ice thickness and  $\tau_b(u, N) = CN^p u^q$ , which is solved together with boundary condition

$$u = U_b \quad \text{at} \quad x = 0, \quad 2A^{-1/n} H \left| \frac{\partial u}{\partial x} \right|^{1/n-1} \frac{\partial u}{\partial x} = \frac{1}{2} \left( 1 - \frac{\rho_i}{\rho_w} \right) H^2 \quad \text{at} \quad x = x_0. \quad (14)$$

All variables and parameters are summarized in the table below, including default values.

|                                |                |                                                                                              |
|--------------------------------|----------------|----------------------------------------------------------------------------------------------|
| Primary variables              | $\phi_c(x, t)$ | Hydraulic potential in channel                                                               |
|                                | $\phi(x, t)$   | Hydraulic potential in sheet                                                                 |
|                                | $S(x, t)$      | Channel cross-sectional area                                                                 |
|                                | $h(x, t)$      | Sheet depth                                                                                  |
|                                | $u(x, t)$      | Ice velocity                                                                                 |
| Derived variables              | $N_c(x, t)$    | Effective pressure in channel                                                                |
|                                | $N(x, t)$      | Effective pressure in sheet                                                                  |
|                                | $\kappa(x, t)$ | Exchange from sheet to channel                                                               |
|                                | $Q(x, t)$      | Discharge in channel                                                                         |
|                                | $q(x, t)$      | Discharge in sheet                                                                           |
|                                | $M(x, t)$      | Melting rate in channel                                                                      |
|                                | $m(x, t)$      | Melting rate in sheet                                                                        |
| Prescribed geometry and inputs | $Z_b(x)$       | Bed elevation [0 m]                                                                          |
|                                | $Z_s(x)$       | Surface elevation                                                                            |
|                                | $m_{in}(x, t)$ | Distributed source to sheet [0 m s <sup>-1</sup> ]                                           |
|                                | $M_{in}(x, t)$ | Distributed source to channel [0 m <sup>2</sup> s <sup>-1</sup> ]                            |
|                                | $x_m$          | Moulin positions [– m]                                                                       |
|                                | $S_m$          | Moulin cross-sectional areas [0 m <sup>2</sup> ]                                             |
|                                | $Q_m(t)$       | Moulin inputs [0 m <sup>3</sup> s <sup>-1</sup> ]                                            |
|                                | $W(x)$         | Width [1000 m]                                                                               |
|                                | $U_b$          | Sliding speed at left boundary [100 m y <sup>-1</sup> ]                                      |
|                                | $\sigma(x)$    | Englacial void fraction connected to sheet [10 <sup>-4</sup> ]                               |
|                                | $\sigma_c(x)$  | Englacial void fraction connected to channel [0]                                             |
| Parameters                     | $\rho_w$       | Density of water [1000 kg m <sup>-3</sup> ]                                                  |
|                                | $\rho_i$       | Density of ice [910 kg m <sup>-3</sup> ]                                                     |
|                                | $g$            | Gravitational acceleration [9.8 m s <sup>-2</sup> ]                                          |
|                                | $L$            | Latent heat of melting [ $3.35 \times 10^5$ J kg <sup>-3</sup> ]                             |
|                                | $c$            | Specific heat capacity of water [4.2 J kg <sup>-1</sup> K <sup>-1</sup> ]                    |
|                                | $\gamma$       | Melting point pressure gradient [ $7.5 \times 10^{-8}$ K Pa <sup>-1</sup> ]                  |
|                                | $G$            | Geothermal heat flux [0.06 W m <sup>-2</sup> ]                                               |
|                                | $n$            | Ice flow law exponent [3]                                                                    |
|                                | $\tilde{A}$    | Ice flow law coefficient in channel [ $5 \times 10^{-24}$ Ps <sup>-3</sup> s <sup>-1</sup> ] |
|                                | $\hat{A}$      | Ice flow law coefficient in sheet [ $5 \times 10^{-24}$ Ps <sup>-3</sup> s <sup>-1</sup> ]   |
|                                | $\alpha_c$     | Channel flux exponent [4/3]                                                                  |
|                                | $K_c$          | Channel flux coefficient [0.05 m s <sup>-1</sup> Pa <sup>-1/2</sup> ]                        |
|                                | $\alpha$       | Sheet flux exponent [3]                                                                      |
|                                | $K$            | Sheet flux coefficient [10 <sup>-6</sup> Pa <sup>-1</sup> s <sup>-1</sup> ]                  |
|                                | $h_r$          | Sheet roughness height [1 m]                                                                 |
|                                | $\ell_r$       | Sheet roughness length [10 m]                                                                |
|                                | $h_{rc}$       | Channel roughness height [0 m]                                                               |
|                                | $\ell_{rc}$    | Channel roughness length [10 m]                                                              |
|                                | $\lambda$      | Exchange coefficient [0 m <sup>2</sup> s <sup>-1</sup> Pa <sup>-1</sup> ]                    |
|                                | $\lambda_2$    | Exchange coefficient [0 m <sup>-1</sup> ]                                                    |
|                                | $\beta$        | Compressibility [ $5 \times 10^{-10}$ Pa <sup>-1</sup> ]                                     |
|                                | $p$            | Sliding law exponent [1]                                                                     |
|                                | $q$            | Sliding law exponent [1]                                                                     |
|                                | $C$            | Sliding law coefficient [ $3 \times 10^5$ m <sup>-1</sup> ]                                  |
|                                | $R$            | Inertial term in momentum equation (artificial) [10 <sup>13</sup> kg m <sup>-3</sup> ]       |

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## Running the model

Download and unzip the files from <http://people.maths.ox.ac.uk/hewitt/wyvis.zip>.

The model is run by typing

```
> [td,ad,ud] = wyvis_solve(td,ad,pd,ud);
```

The inputs are:

**td** a row vector of times at which the output is requested (the first and last entries indicate the time to integrate over).

**ad** a struct containing the prescribed geometry and inputs.

**pd** a struct containing the parameters.

**ud** a struct containing the initial conditions.

The outputs are:

**td** a row vector of times at which output has been stored.

**ad** a struct containing the prescribed geometry and inputs (with any fields missing in the input having been filled in).

**ud** a 1-by-L struct containing the solution at each time ( $L=\text{length}(\text{td})$ ).

The last input **ud** is optional; if it is not given, an initial condition will be assigned automatically and this will typically lead to significant transient behaviour. Initial conditions that are a long way from the incompressible limit of the equations will likely cause trouble and may cause the integration to fail.

Default values for the parameters in **pd** can be assigned by typing

```
>pd = wyvis_default_parameters,
```

Default values for most fields of **ad** will be assigned automatically if they are missing, but at the least the fields **x**, **Z\_b** and **Z\_s** must be assigned. These should define the grid points, the bed elevation, and the surface elevation, respectively, as column vectors. For example, typing

```
>ad.x = linspace(0,10000,40)';
```

```
>ad.Z_b = 0*ad.x;
```

```
>ad.Z_s = 0.1*(ad.x(end)-ad.x);
```

will assign 40 grid points between 0 and 10 km, with a flat bed and a downward sloping surface with a constant slope of 0.1. By default, there are no moulins and all water inputs are zero. The model will not run like this as some form of input must be provided. Water inputs should be given as function handles, using  $@(t)$ . For example,

```
>ad.M_in = @(t) [ones(size(ad.x))]*1e-3*t.^0;
```

assigns a constant source of  $10^{-3} \text{ m}^2 \text{ s}^{-1}$  to the channel. (This makes **ad.M\_in** a function of **t**, so that typing **>ad.M\_in(0)** or **>ad.M\_in(60)** returns the input at  $t = 0$  or  $t = 60$  s, for example; in this case both would return a column vector of length 40 (*i.e.* **size(ad.x)**) with all entries  $10^{-3}$ ).

If an initial condition **ud** is to be given, this struct should contain the fields **S**, **h**, **phi\_c** and **phi**, and **u** as column vectors. The output solution struct **ud** will contain additional fields for derived variables such as the effective pressure and flux (type **>ud(end)** to see a list).

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## Example

Following the above, a simple run to a steady state can be achieved by typing

```
>pd = wyvis_default_parameters,  
>ad.x = linspace(0,10000,40)';  
>ad.Z_b = 0*ad.x;  
>ad.Z_s = 0.1*(ad.x(end)-ad.x);  
>ad.M_in = @(t) [ones(size(ad.x))]*1e-3*t.^0;  
>[td,ad,ud] = wyvis_solve( [0:1:365]*24*60*60 , ad , pd );  
>wyvis_plot_all_c( 1 , ad , ud(end) );
```

These commands can be found in the m-file `example.m`. They will run the model for one year, saving output at daily time steps. The last command plots the solution at the final time step in figure 1 (see below). Hopefully by that time a steady state has been achieved. The evolution to a steady state can be checked by plotting the values of `phi_c` over time, for example:

```
>figure(2); plot( td , [ud.phi_c] );
```

Having achieved a more or less steady state, this can then be used as a reasonable initial condition to consider a time varying water input, for example:

```
>ad.M_in = @(t) [ones(size(ad.x))]*1e-3*( 1 + 0.1*sin(2*pi*t/24/60/60) );  
>[td,ad,ud] = wyvis_solve( [0:1/24:10]*24*60*60 , ad , pd , ud(end) );  
>wyvis_plot_phi_ct( 1 , td , ud , 20 );  
>wyvis_movie_phi_c( 2 , ad , ud , td );
```

The first command here modifies the water input so that it has a 10% sinusoidal oscillation with period of 1 day. The second command runs the model for 10 days, outputting at hourly time steps, and using the final state from the previous run as the initial condition (`ud(end)`). The first plot in figure 1 produces a time series of the solution at the 20th grid point. The ‘movie’ in figure 2 shows the evolution of the hydraulic potential.

## Plotting commands

It is straightforward to plot parts of the solution by typing, for example,

```
>plot( ad.x , ud(1).S );
```

to see the channel cross-sectional area  $S$  as a function of  $x$  at the first time step. There are also a few specific commands to help plot certain things more quickly. In each case `fig` is the figure number to plot in. The grid indices `xi` are optional; all grid points will be plotted if this is not included. Commands with `_c` in the name plot only the channel variables.

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|                                                   |                                                                                   |
|---------------------------------------------------|-----------------------------------------------------------------------------------|
| <code>&gt;wyvis_plot_phi_c(fig,ad,ud);</code>     | Plot $\phi_c$ against $x$                                                         |
| <code>&gt;wyvis_plot_all_c(fig,ad,ud);</code>     | Plot all channel variables against $x$                                            |
| <code>&gt;wyvis_plot_phi_ct(fig,td,ud,xi);</code> | Plot time series of $\phi_c$ against $t$ for the $x$ ith grid points              |
| <code>&gt;wyvis_plot_all_ct(fig,td,ud,xi);</code> | Plot time series of all channel variables against $t$ for the $x$ ith grid points |
| <code>&gt;wyvis_movie_phi_c(fig,ad,ud,td);</code> | Plot $\phi_c$ against $x$ for each time step sequentially                         |
| <code>&gt;wyvis_plot_phi(fig,ad,ud);</code>       | Plot $\phi_c$ and $\phi$ against $x$                                              |
| <code>&gt;wyvis_plot_all(fig,ad,ud);</code>       | Plot all variables against $x$                                                    |
| <code>&gt;wyvis_plot_phit(fig,td,ud,xi);</code>   | Plot time series of $\phi_c$ and $\phi$ against $t$ for the $x$ ith grid points   |
| <code>&gt;wyvis_plot_allt(fig,td,ud,xi);</code>   | Plot time series of all variables against $t$ for the $x$ ith grid points         |
| <code>&gt;wyvis_movie_phi(fig,ad,ud,td);</code>   | Plot $\phi_c$ and $\phi$ against $x$ for each time step sequentially              |
| <code>&gt;wyvis_plot_geometry(fig,ad);</code>     | Plot bed and surface elevations                                                   |
| <code>&gt;wyvis_plot_u(fig,ad,ud);</code>         | Plot ice velocity against $x$                                                     |
| <code>&gt;wyvis_plot_u(fig,ad,ud);</code>         | Plot time series of ice velocity against $t$ for the $x$ ith grid points          |